Akoizo Original Practice - Form 4

Division B | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which material is NOT allowed in a Boomilever?

Unit: Wood selection, density, and buckling.
Worked-case data: A strut buckles at 120 N with an unbraced length of 30 cm. A brace is added at its midpoint. Estimate the new buckling load.
Separate investigation observations:
Stick A: 0.11 g/cm^3 - failed as a strut at 6 kg.
Stick B: 0.18 g/cm^3, same size - held 11 kg as a strut.
Stick C: 0.11 g/cm^3 with a midpoint brace - held 15 kg.

A. Basswood

B. Hard balsa

C. Commercial plywood

D. Team-laminated balsa

Answer: _____

2. (2 points) How do compression members usually fail in a Boomilever?

Unit: Statics of a cantilever: moments, tension, and compression.
Worked-case data: A 15 kg load hangs 42 cm from the wall. The strut meets the wall 20 cm below the bolt. Estimate the horizontal chord force.
Separate investigation observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. By crushing the bolt

B. By buckling sideways

C. By stretching apart

D. By melting the glue

Answer: _____

3. (2 points) A joint failed and glue remained on only one surface. What does that suggest?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. Poor bond area or fit at that joint

B. The wood was too dense

C. The load was too far out

D. The bolt was loose

Answer: _____

4. (2 points) Why do soft-wood bolt hooks often fail by crushing?

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. The round bolt concentrates force on a small area of soft wood

B. The bolt heats up under load

C. Glue cannot touch metal

D. The bolt is too far from the wall

Answer: _____

5. (4 points) Explain why the bonus contact line usually makes a Boomilever heavier.

Answer: _____
_____
_____

6. (2 points) Which prototype should the team bring, based on expected score?

Supplied synthetic observations:
Prototype 1 (base): 9.6 g, held 15,000 g, contact at 14 cm below the bolt.
Prototype 2 (bonus): 12.1 g, held 15,000 g, contact above the 15 cm line.
Prototype 3 (bonus, lightened): 10.4 g, failed at 12,300 g when a joint cracked.
Supervisor note: Prototype 1's contact point is above the 20 cm line.

A. Prototype 3

B. Prototype 2

C. Prototype 1

Answer: _____

7. (4 points) Why is pouring sand steadily better than dumping it quickly?

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

Answer: _____
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_____

8. (4 points) Give one reason the tension chord often fails at its ends rather than its middle.

Answer: _____
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9. (2 points) What do these tests show about where the structure needs material?

Supplied synthetic observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. All members should be the same size

B. The strut needs bracing and joints need care; the chord is already strong enough

C. The chord needs to be much thicker

Answer: _____

10. (4 points) How does humidity affect a finished structure?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

Answer: _____
_____
_____

11. (6 points) Solve this case and explain your reasoning: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?

Answer: _____
_____
_____

12. (2 points) Define “Contact depth” in the context of Boomilever.

Answer: _____
_____
_____

13. (2 points) Define “Loading Assembly” in the context of Boomilever.

Answer: _____
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_____

14. (2 points) Define “Testing Wall” in the context of Boomilever.

Answer: _____
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15. (2 points) Define “Modulus of elasticity (E)” in the context of Boomilever.

Answer: _____
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16. (2 points) Define “Pyramid layout” in the context of Boomilever.

Answer: _____
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17. (2 points) Define “Free-body diagram” in the context of Boomilever.

Answer: _____
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18. (2 points) Define “Safety factor” in the context of Boomilever.

Answer: _____
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